Topic quiz: Forces

IBDP Physics SL · Grade 11 · A.2 Forces · Newton's laws · 10 multiple-choice questions

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Name: Class: Date:

How to use. Practise with the simulations and worksheets first. Then answer every question: circle one letter and write it in the grid below. Use g = 9.81 m s^{-2} and show any working next to the question.
Check yourself online: the same quiz with instant feedback and explanations — https://school.taneski.com/physics/dp/quiz-forces/

1	2	3	4	5	6	7	8	9	10

Question 1

A book rests on a table. According to Newton's third law, which force is the partner of the book's weight?

(A) The normal force of the table on the book
(B) The push of the book on the table
(C) The gravitational pull of the book on the Earth
(D) The friction of the table on the book

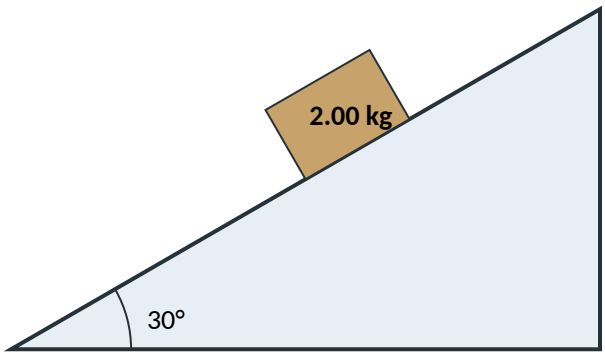
Question 2

A 60.0 kg student stands on bathroom scales in a lift that is accelerating upwards at 1.50 m s^{-2}. What do the scales read?

(A) 90 N
(B) 499 N
(C) 589 N
(D) 679 N

Question 3

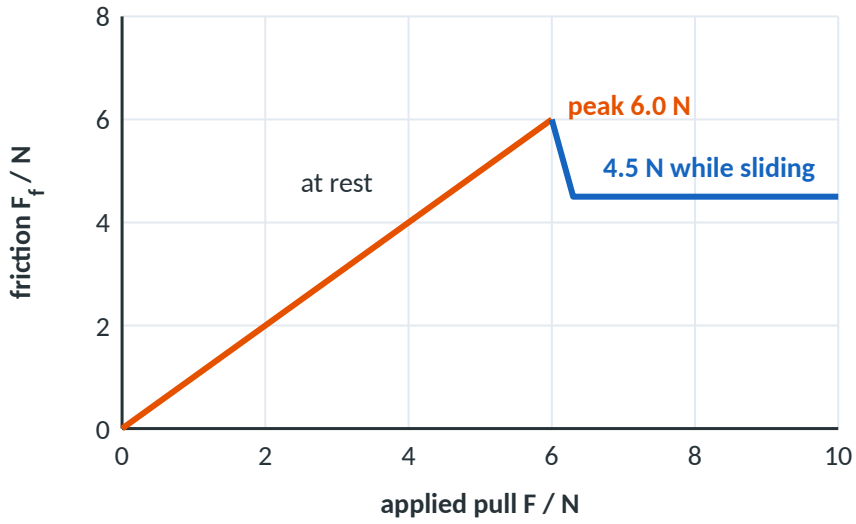
A 2.00 kg block rests on a ramp at 30° to the horizontal. What is the normal force on the block?



(A) 9.81 N
(B) 17.0 N
(C) 19.6 N
(D) 22.7 N

Question 4

A block of weight 15.0 N is pulled along a horizontal bench with a slowly increasing force. The graph shows the friction force. What are the coefficients of static and dynamic friction?



(A) $\mu_s = 0.30$ and $\mu_d = 0.40$
(B) $\mu_s = 0.40$ and $\mu_d = 0.40$
(C) $\mu_s = 0.40$ and $\mu_d = 0.30$
(D) $\mu_s = 2.5$ and $\mu_d = 3.3$

Question 5

A ramp is tilted slowly. A block on it just starts to slip when the angle is 28°. What is the coefficient of static friction?

(A) 0.47
(B) 0.53
(C) 0.88
(D) 1.9

Question 6

The block from the last question is replaced by a heavier block of the same material. At what angle does the heavier block start to slip?

(A) At a smaller angle, because its weight down the slope is larger

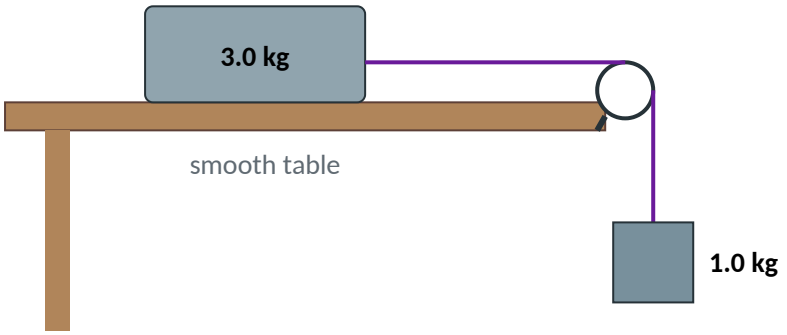
(B) At a larger angle, because the friction force is larger

(C) At the same angle, because the mass cancels out

(D) It never slips

Question 7

A 3.0 kg cart on a smooth table is pulled by a string over a pulley by a hanging 1.0 kg mass. What is the tension in the string while the masses accelerate?



(A) 2.45 N

(B) 7.36 N

(C) 9.81 N

(D) 29.4 N

Question 8

A spring stretches by 4.0 cm when a 0.20 kg mass hangs from it at rest. What is the spring constant?

(A) 0.049 N m^{-1}

(B) 0.49 N m^{-1}

(C) 5.0 N m^{-1}

(D) 49 N m^{-1}

Question 9

A steel ball falls through glycerol at its terminal velocity. W is its weight, F_d the drag and F_b the buoyant force. Which equation is correct?

(A) $W = F_d + F_b$

(B) $F_d = W + F_b$

(C) $F_d = 0$

(D) $W > F_d + F_b$

Question 10

An object weighs 2.50 N in air and 1.88 N when fully under water (ρ_{water} = 1000 kg m^{-3}). What is its volume?

(A) 63 cm^3

(B) 192 cm^3

(C) 255 cm^3

(D) 620 cm^3

End of quiz · Score: _____ / 10